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## Artificial Intelligence Driven Robotics

In the past three decades, technology has improved significantly, and one of the many fields that has continued to innovate and expand is artificial intelligence driven robotics.

Artificial intelligence(AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. AI-driven robotics is the field of robotics that uses artificial intelligence and machine learning to build and program robots to automate or carry out tasks. With companies like Amazon and Google leading the way in the field, **artificial intelligence driven robotics is going to bring a change to society's workforce.**

### WHAT IS AI-DRIVEN ROBOTICS

Before addressing the effects AI-driven robotics will have on society's workforce, it is important to understand what AI is and how it is being used in the field of robotics. The goal of using AI-driven robotics is to reduce the amount of human labor needed to complete a task.

More companies are starting to find ways to replace human jobs with robots, in efforts to reduce the cost of labor and increase productivity.

### HOW AI-DRIVEN ROBOTICS IS AFFECTING SOCIETY'S WORKFORCE

One company that demonstrates this change in labor practices is Amazon. Amazon, the second-largest employer in the United States, is starting to replace human labor with robot labor. In the period of 2017-2020, Amazon went from having less than fifty-thousand robots in their workforce to having over two million in 2020 ("Amazon Robots"). Given that Amazon's

workforce has an estimated one million human workers (“Amazon Robots”), this data estimates that two-thirds of Amazon’s workforce is made up of robots.

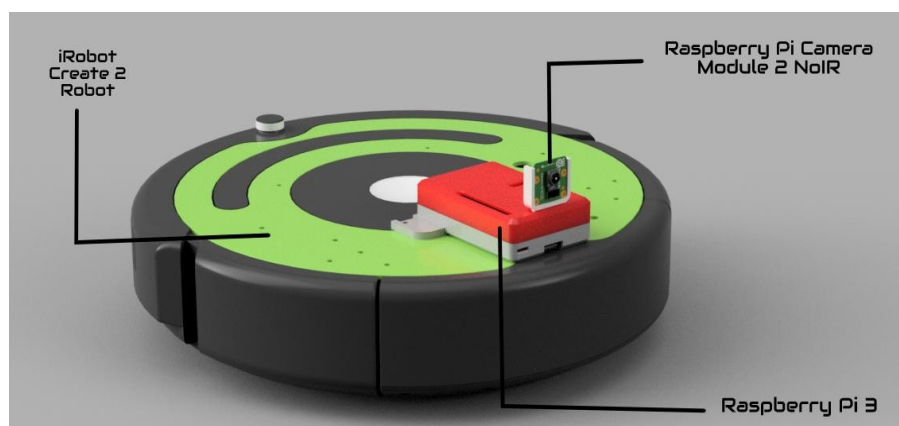
More money is being put into AI-driven robotics, which is leading to more use of robots, more production of robots, and more research into the use of AI-driven robots in the workforce. According to a study conducted by Precedence Research, the AI-robotics market is projected to increase by 484 percent in the next eight years, with a total evaluation of 54.3 billion dollars by the year 2030.

Within the past three decades, the cost of human labor has increased while the cost of robot labor has decreased. A study by the Economist Intelligence Unit found that robot prices in the United States have dropped by over forty percent since 1990, while labor costs have increased by over one hundred twenty percent (“Cost of Automation”). This trend is reflected in the data, as the average number of robots per thousand workers in the United States increased from 0.4 in 1995 to 1.6 in 2015 (“Robots and Jobs” 41).

## PRODUCT OVERVIEW

This product is called the Tracking Robot using Artificial Intelligence Navigation (TRAIN). TRAIN was designed to track objects using artificial intelligence and navigate toward them autonomously.

## EARLY DESIGN



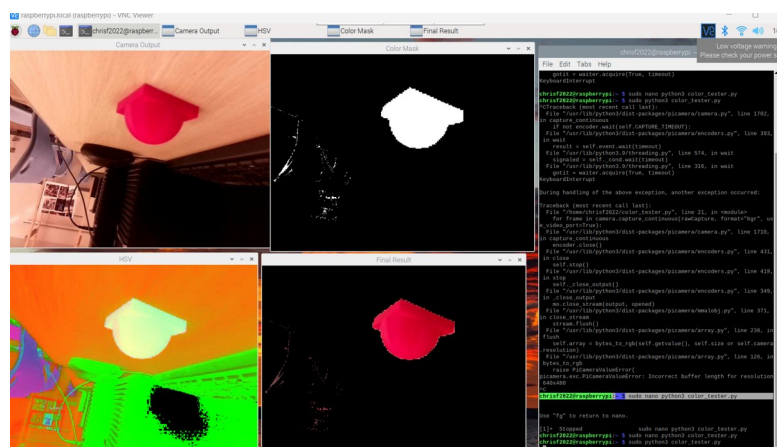
## Early render of TRAIN (First Design, November 22, 2022)

The image shown above is the first full rendering of TRAIN. TRAIN uses an iRobot Create 2 robot as a base. This robot was chosen as a base for TRAIN because the Create 2 is designed to be programmed in Python, and is interfaceable with the Raspberry Pi. This product was designed with the goal of tracking objects using artificial intelligence and autonomously moving toward those objects. This robot is an example of a robot that could be used in the workforce and has the potential to replace a human job.

## HARDWARE AND SOFTWARE

As labeled in the render of TRAIN, this robot makes use of a Raspberry Pi 3 and a Raspberry Pi Camera Module 2 NoIR. The Raspberry Pi is a credit card-sized mini-computer that can be used for many small computing tasks. For this product, the Raspberry Pi was used to program and run the AI responsible for object tracking and autonomous navigation. The Raspberry Pi Camera Module 2 NoIR is an eight-megapixel camera capable of taking infrared photographs of 3280 x 2464 pixels. This camera allows for the detection of objects and translates the position of the objects into data values that are processed through the Raspberry Pi and returned to the iRobot Create 2.

Initially, this product used color recognition to track objects. Using color recognition to track objects came with many problems. Shown below is the first working model of the red color tracking.



As seen in the image above, there are two factors that are undesirable for this product. Firstly, the camera being used for the product had trouble isolating the hue of red it was supposed to be tracking and required an external light source to track to object. The second problem was that the camera was receiving red color data from unwanted sources, such as lighting or from a person's skin. Both of these issues led to instances where the camera was unable to properly recognize the color red. To add to this, the way the camera receives color information is inconsistent. If the lighting in the room is changed, even slightly, the color of the object will be a completely different value. This means that due to hardware limitations and financial constraints, color tracking was not a viable method for object tracking. Because of this, the product was changed to track objects through a machine-learning library called TensorFlow.

#### WHAT IS TENSORFLOW

TensorFlow is an open-source library developed by Google primarily for deep learning applications. TensorFlow was originally developed for large numerical computations without keeping deep learning in mind, however it has also been used to help train and build machine learning models.

#### FIRST PROTOTYPE

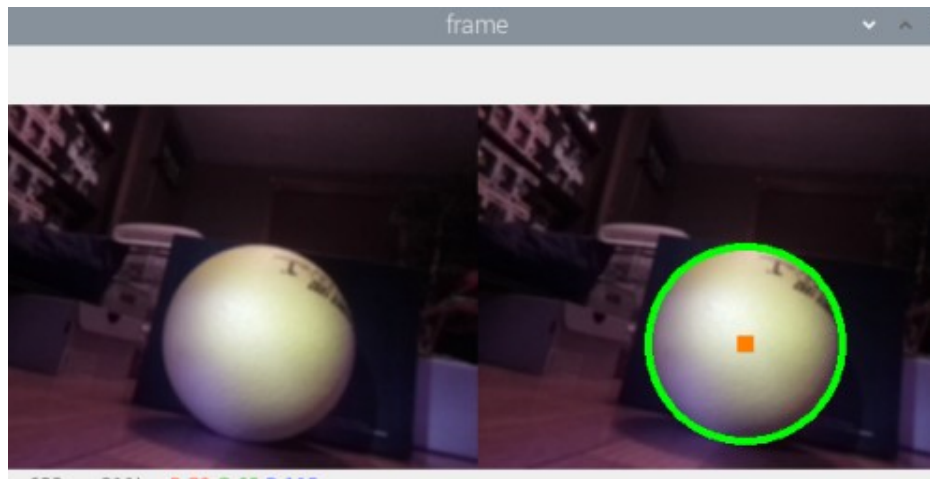


First prototype of TRAIN (December 7, 2022)

After completing the initial design phase and assembling all parts of the product, a proof of concept prototype of TRAIN was finished on December 7, 2022. At this point in the design process, the programming for both object tracking and autonomous navigation was incomplete. This prototype mainly served the purpose of testing the mounting position for the Raspberry Pi and ensuring the camera would be able to stay upright.

## IMPROVEMENTS AND ALTERATIONS

While testing object tracking during the prototyping phase, it was made apparent that the camera being used was returning an upside-down video output, which was altering the data being sent to the robot. To fix this, a line of code was added to rotate the camera output one hundred eighty degrees, as shown in the image below.

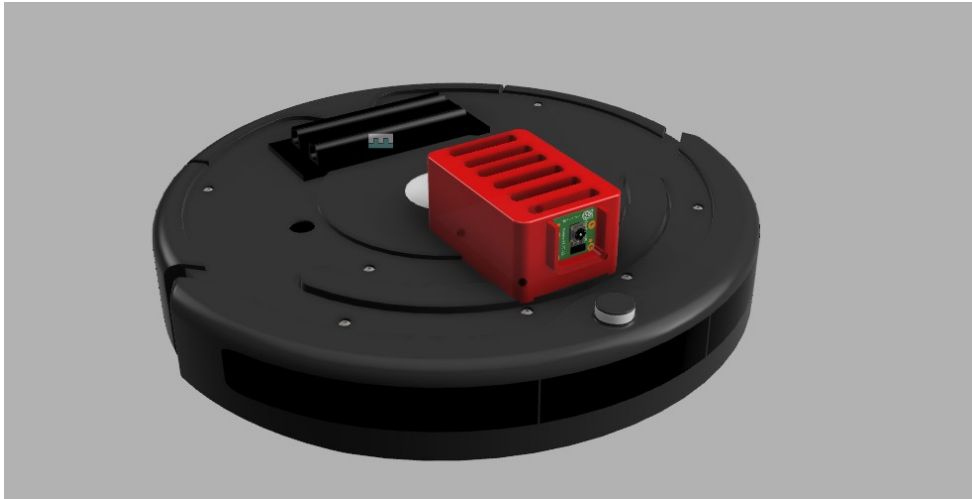


Screenshot of TensorFlow tracking program (December 29, 2022)

When designing this product, the processing capabilities of the Raspberry Pi were vastly overestimated. While testing the object tracking program, the camera was returning video at a rate of two frames per second. This is significantly slower than the desired framerate. Measures were taken in both the program and the execution of the design to increase framerate, in order to

get more information per second, including compressing the length of the program and reducing the number of objects the program can track.

## FINAL PRODUCT



Final render of TRAIN (January 3, 2023)

A PiJuice portable battery was added to the final version of TRAIN to allow the robot to move freely without needing to be plugged into a wall or computer. The finished version of TRAIN was completed on December 29, 2022. TRAIN can find, track, and navigate toward objects completely autonomously. Because the battery added height, a new case was made to account for the change in height. The newer case features a more secure camera mount and wire routing holes on top of the case. Tape reinforcements and a 3D-printed wire mount were added to make the product more stable and reduce the risk of wires getting caught or parts falling off.

## CONCLUSION

TRAIN was designed with the goal of demonstrating one way that AI-driven robotics could change the workforce. The use of AI-driven robotics in the workforce is already starting to bring about changes in work environments across the world. As technology continues to advance, it is likely that the impact of AI-driven robotics on the workforce will only increase.

This product is just one example of the many ways that AI-driven robotics has the potential to transform society's workforce.

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